

ABDULLAH AL OTAIBI

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EDUCATION

The Pennsylvania State University, University Park

Expected Graduation: December 2026

Bachelor of Science in Mechanical Engineering

Cumulative GPA: 3.84 / 4.00 | Major GPA: 4.00 / 4.00

Relevant Coursework: Mechatronics (ME 454), Vibration of Mechanical Systems (ME 370), Design Methodology (ME 340), Product Design & Manufacturing Processes (IE 312), Thermodynamics, Fluid Mechanics, Heat Transfer, Dynamic Systems

HONORS & AWARDS

- Aramco CCP Scholarship Recipient (Oct 2024 – Present) — Awarded through Saudi Aramco's highly competitive College Continuation Program to top Saudi students worldwide; full undergraduate sponsorship with guaranteed engineering placement upon graduation.
- Tau Beta Pi Engineering Honor Society — Initiated Spring 2026
- President Walker Award — Awarded for cumulative GPA of 4.00 in 12–36 graded Penn State credits

ENGINEERING PROJECTS

Nittany Lion Light Stand — Arduino Desk Lamp

Personal Project

- Designed and 3D-printed a lion-shaped lamp housing; integrated an Arduino-controlled servo arm for dynamic repositioning of the light source.
- Programmed servo motion sequences and LED control logic in Arduino C++; assembled full electromechanical system from hardware selection through final calibration.

Autonomous Wall-Following Robot

ME 454 Mechatronics

- Implemented PD and finite-state-machine controllers in Arduino C++ on a RedBot platform using IR sensor feedback for closed-loop wall tracking.
- Tuned controller gains across multiple hardware units; diagnosed and resolved motor wiring asymmetry through systematic debugging.

Pendulum Tuned Mass Damper Design

ME 370 Vibration of Mechanical Systems

- Derived 4-DOF nonlinear equations of motion via Euler-Lagrange formulation for a three-story shear building with a PTMD; linearized via small-angle approximation.
- Implemented modal superposition in MATLAB and ran a parametric grid search to identify optimal PTMD parameters meeting a 7 cm peak-displacement constraint under impulse loading.

e-Formula Car Body Aerodynamic Optimization

ME 340 Design Methodology

- Modeled three iterative car designs in SolidWorks (baseline, Alpha-1, Alpha-2) targeting reduced drag at sustained high-speed lap conditions.
- Validated drag and lift performance through wind tunnel testing and CFD analysis; correlated results to predicted lap-time improvement.
- Competed against 64 teams in the Penn State e-Formula design competition, advancing to the quarterfinals.

PROFESSIONAL EXPERIENCE

AI-Driven Materials & Manufacturing Program

May 2026 – June 2026

- Completed Phase 1 machine learning training as part of Prof. Hongtao Sun's competitive undergraduate research program.

Penn State Department of Economics | Economics Grader

Aug 2024 – Dec 2024

- Graded homework and exams for ECON 102 under Prof. Dave Brown, ensuring fairness and consistency in evaluation.
- Provided written feedback to students and proctored exams alongside the instructor.

LEADERSHIP & ACTIVITIES

Penn State Table Tennis Club | Secretary & Event Planner

Aug 2024 – Feb 2025

- Coordinated with Penn State Competitive Sports staff to organize community events and recruitment drives.
- Promoted the club at the Fall 2024 Involvement Fair and represented Penn State at regional tournaments.

SKILLS

Software & Programming: SolidWorks, MATLAB / Simulink, Arduino IDE, C / C++, Microsoft Excel

Engineering Methods: Closed-loop control (PD, PID, FSM), embedded systems, CFD, modal analysis

Languages: English (fluent), Arabic (native)